

Wireshark Filter - Most Important Problems for Exam

40 Essential Problems You MUST Know

⚠️ READ THIS FIRST - Critical Differences

Capture Filter vs Display Filter Syntax:

Purpose	Capture Filter	Display Filter
IP Address	host 10.0.0.1	ip.addr==10.0.0.1
Source IP	src host 10.0.0.1	ip.src==10.0.0.1
Port	port 80	tcp.port==80
Network	net 192.168.1.0/24	ip.addr==192.168.1.0/24
Protocol	tcp	tcp
Equality	= (or none)	== (double equals)
AND operator	and	and (or &&)
OR operator	or	or (or)
NOT operator	not	! (or not)

REMEMBER:

- Capture filters = BPF syntax (used DURING capture)
- Display filters = Wireshark syntax (used AFTER capture)

Part 1: CAPTURE FILTERS (20 Problems)

These run DURING packet capture - cannot be changed later!

● PROBLEM 1: Single Host (Most Basic)

Question: Capture all traffic from/to IP 192.168.1.100

Capture Filter:

host 192.168.1.100

Explanation:

- Most common filter
- Captures bidirectional traffic (both incoming and outgoing)
- Works for both source AND destination

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● PROBLEM 2: Source IP Only

Question: Capture only traffic FROM 10.0.0.50

Capture Filter:

```
src host 10.0.0.50
```

Explanation:

- `src` = source only
- Captures outgoing traffic from this IP
- Does NOT capture incoming responses

Exam Frequency: ★ ★ ★ ★ ★ (High)

● PROBLEM 3: Destination IP Only

Question: Capture only traffic going TO server 172.16.5.200

Capture Filter:

```
dst host 172.16.5.200
```

Explanation:

- `dst` = destination only
- Captures incoming traffic to this IP
- Useful for monitoring server load

Exam Frequency: ★ ★ ★ ★ ★ (High)

● PROBLEM 4: Two Hosts Communication

Question: Capture traffic between 10.0.0.5 and 10.0.0.10 ONLY

Capture Filter:

host 10.0.0.5 and host 10.0.0.10

Explanation:

- `and` means BOTH must be present
- One is source, other is destination
- Bidirectional conversation
- Excludes all other traffic

Exam Frequency: ★★☆☆☆ (Very High)

● PROBLEM 5: Single Port (HTTP)

Question: Capture all HTTP traffic (port 80)

Capture Filter:

tcp port 80

OR

port 80

Explanation:

- `tcp port 80` more specific
- `port 80` works for both TCP and UDP
- Captures both source and destination port 80

Exam Frequency: ★★☆☆☆ (Very High)

● PROBLEM 6: DNS Queries (Destination Port)

Question: Capture only DNS queries (TO server, not responses)

Capture Filter:

udp dst port 53

Explanation:

- `dst port 53` = going TO DNS server (queries)
- `src port 53` = coming FROM server (responses)
- DNS uses UDP on port 53

Exam Frequency: ★ ★ ★ ★ ★ (Very High - Very Tricky!)

● PROBLEM 7: Multiple Ports (Web Traffic)

Question: Capture HTTP and HTTPS traffic

Capture Filter:

tcp port 80 or tcp port 443

Explanation:

- `or` means EITHER condition
- Port 80 = HTTP, Port 443 = HTTPS
- Parentheses not needed for simple OR

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● PROBLEM 8: Exclude Traffic (NOT)

Question: Capture all traffic EXCEPT SSH (port 22)

Capture Filter:

not tcp port 22

Explanation:

- `not` excludes specified traffic
- Removes administrative/encrypted traffic
- Reduces capture noise

Exam Frequency: ★ ★ ★ ★ ★ (High)

● PROBLEM 9: Subnet Filtering

Question: Capture traffic from network 192.168.1.0/24

Capture Filter:

```
net 192.168.1.0/24
```

Explanation:

- `net` keyword for network ranges
- `/24` = subnet mask 255.255.255.0
- Captures 192.168.1.1 to 192.168.1.254

Exam Frequency: ★★★★★ (Very High)

● PROBLEM 10: Exclude Broadcast/Multicast

Question: Capture unicast traffic only (no broadcast/multicast)

Capture Filter:

```
not broadcast and not multicast
```

Explanation:

- Removes network noise
- `and` means BOTH conditions must be true
- Very common in production captures

Exam Frequency: ★★★★★ (High)

● PROBLEM 11: Host + Port Combination

Question: Capture HTTP traffic to/from server 10.20.30.40

Capture Filter:

```
host 10.20.30.40 and tcp port 80
```

Explanation:

- Combines host and port filters
- Both conditions must be true

- Specific service on specific host

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● **PROBLEM 12: Host + Multiple Ports (CRITICAL!)**

Question: Capture web traffic (HTTP/HTTPS) to server 192.168.1.100

Capture Filter:

host 192.168.1.100 and (tcp port 80 or tcp port 443)

Explanation:

- **PARENTHESES ARE CRITICAL!**
- Without parentheses: WRONG LOGIC
- Groups OR conditions properly

Wrong Answer:

host 192.168.1.100 and tcp port 80 or tcp port 443

This means: (host AND port 80) OR (port 443 from anywhere)

Exam Frequency: ★ ★ ★ ★ ★ (Very High - Common Mistake!)

● **PROBLEM 13: Protocol Filtering (TCP)**

Question: Capture only TCP packets

Capture Filter:

tcp

Explanation:

- Simple protocol name
- Excludes UDP, ICMP, etc.
- Most common transport protocol

Exam Frequency: ★ ★ ★ ★ ★ (High)

● PROBLEM 14: ICMP (Ping)

Question: Capture only ICMP packets

Capture Filter:

```
icmp
```

Explanation:

- ICMP = ping, traceroute, errors
- Network diagnostic protocol
- No port numbers (Layer 3)

Exam Frequency: ★★☆☆ (High)

● PROBLEM 15: Network to Host

Question: Capture traffic FROM subnet 172.16.0.0/24 TO server 10.0.0.100

Capture Filter:

```
src net 172.16.0.0/24 and dst host 10.0.0.100
```

Explanation:

- Directional filter
- Source is network, destination is host
- One-way traffic only

Exam Frequency: ★★☆☆ (High)

● PROBLEM 16: TCP SYN Packets (Port Scan Detection)

Question: Capture TCP SYN packets (connection attempts)

Capture Filter:

```
tcp[tcpflags] & tcp-syn != 0 and tcp[tcpflags] & tcp-ack == 0
```

OR (simpler):

```
tcp[13] & 2 != 0 and tcp[13] & 16 == 0
```

Explanation:

- `tcp[13]` = TCP flags byte
- `& 2` checks SYN flag (value 2)
- `& 16 == 0` ensures ACK is NOT set
- Pure SYN = connection initiation/port scan

TCP Flag Values:

SYN = 2
ACK = 16
FIN = 1
RST = 4

Exam Frequency: ★ ★ ★ ★ ★ (Very High - Advanced)

● PROBLEM 17: Multiple Exclusions

Question: Capture all traffic EXCEPT HTTP and HTTPS

Capture Filter:

`not (tcp port 80 or tcp port 443)`

Explanation:

- Parentheses group OR conditions
- NOT applied to entire group
- Without parentheses: WRONG!

Wrong:

`not tcp port 80 or tcp port 443`

Means: (NOT port 80) OR (port 443) - includes 443!

Exam Frequency: ★ ★ ★ ★ ★ (Very High - Common Mistake!)

● PROBLEM 18: Email Traffic (Multiple Protocols)

Question: Capture email traffic (SMTP, POP3, IMAP)

Capture Filter:

tcp port 25 or tcp port 110 or tcp port 143

Explanation:

- Port 25 = SMTP (sending)
- Port 110 = POP3 (receiving)
- Port 143 = IMAP (accessing)
- All use TCP

Exam Frequency: ★★☆☆ (High)

● **PROBLEM 19: MAC Address**

Question: Capture traffic from MAC address 00:11:22:33:44:55

Capture Filter:

ether host 00:11:22:33:44:55

Explanation:

- ether = Ethernet layer (Layer 2)
- Works before IP is assigned (DHCP)
- Source OR destination MAC

Exam Frequency: ★★★ (Medium)

● **PROBLEM 20: Complex Scenario**

Question: Capture HTTP/HTTPS from subnet 192.168.1.0/24 to server 10.0.0.100

Capture Filter:

src net 192.168.1.0/24 and dst host 10.0.0.100 and (tcp port 80 or tcp port 443)

Explanation:

- Three conditions combined
- Source network + Destination host + Ports
- Parentheses for port OR condition
- Most complex type in exams

Exam Frequency: ★★☆☆☆ (Very High - Final Boss!)

Part 2: DISPLAY FILTERS (20 Problems)

These run AFTER capture - can be changed anytime!

● PROBLEM 21: Source IP

Question: Show packets from source IP 192.168.1.50

Display Filter:

```
ip.src==192.168.1.50
```

Explanation:

- `ip.src` = source IP field
- **Double equals** == (not single!)
- Display filter syntax

Exam Frequency: ★★☆☆☆ (Very High)

● PROBLEM 22: Either Direction IP

Question: Show packets involving IP 10.0.0.25 (any direction)

Display Filter:

```
ip.addr==10.0.0.25
```

Explanation:

- `ip.addr` = source OR destination
- Most commonly used
- Bidirectional traffic

Exam Frequency: ★★☆☆☆ (Very High)

● PROBLEM 23: Destination Port

Question: Show packets going to port 443 (HTTPS)

Display Filter:

```
tcp.dstport==443
```

OR

```
tcp.port==443
```

Explanation:

- `tcp.dstport` = destination port (more specific)
- `tcp.port` = source OR destination port

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● PROBLEM 24: HTTP Protocol

Question: Show only HTTP packets

Display Filter:

```
http
```

Explanation:

- Simple protocol name
- Case insensitive
- Shows complete HTTP sessions

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● PROBLEM 25: HTTP GET Requests

Question: Show only HTTP GET requests

Display Filter:

```
http.request.method=="GET"
```

Explanation:

- `http.request.method` = HTTP method field
- Use quotes for string values
- Shows which pages users requested

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● PROBLEM 26: HTTP Response Code

Question: Show HTTP 404 Not Found errors

Display Filter:

```
http.response.code==404
```

Explanation:

- `http.response.code` = status code
- 404 = page not found
- Only in responses, not requests

Common Codes:

```
200 = OK
301/302 = Redirect
403 = Forbidden
404 = Not Found
500 = Server Error
```

Exam Frequency: ★ ★ ★ ★ ★ (Very High)

● PROBLEM 27: TCP SYN Flag

Question: Show TCP packets with SYN flag set (connection requests)

Display Filter:

```
tcp.flags.syn==1
```

Explanation:

- `tcp.flags.syn` = SYN flag
- `==1` means flag is SET

- Connection initiation packets

Exam Frequency: ★★☆☆☆ (Very High)

● PROBLEM 28: TCP SYN Only (Pure SYN)

Question: Show only pure SYN packets (SYN=1, ACK=0)

Display Filter:

```
tcp.flags.syn==1 and tcp.flags.ack==0
```

OR

```
tcp.flags==0x02
```

Explanation:

- First method: explicit flags
- Second method: hex value (0x02 = SYN only)
- First step of 3-way handshake
- Port scan detection

Exam Frequency: ★★☆☆☆ (Very High - CRITICAL!)

● PROBLEM 29: TCP SYN-ACK

Question: Show TCP SYN-ACK packets (server responses)

Display Filter:

```
tcp.flags.syn==1 and tcp.flags.ack==1
```

OR

```
tcp.flags==0x12
```

Explanation:

- Both SYN and ACK flags set
- 0x12 hex = 18 decimal
- Second step of 3-way handshake
- Server accepting connection

Exam Frequency: ★★★★★ (Very High - CRITICAL!)

● PROBLEM 30: TCP Retransmissions

Question: Show all TCP retransmitted packets

Display Filter:

```
tcp.analysis.retransmission
```

Explanation:

- Wireshark auto-detects retransmissions
- No `==1` needed (field existence = true)
- Indicates packet loss/network issues

Exam Frequency: ★★★★★ (High)

● PROBLEM 31: DNS Queries

Question: Show only DNS queries (not responses)

Display Filter:

```
dns.flags.response==0
```

Explanation:

- `response==0` = query
- `response==1` = response
- Shows what domains are being looked up

Exam Frequency: ★★★★★ (Very High - Tricky!)

● PROBLEM 32: DNS Query Name

Question: Show DNS lookups for "google.com"

Display Filter:

```
dns.qry.name=="google.com"
```

OR (partial match):

```
dns.qry.name contains "google.com"
```

Explanation:

- First: exact match
- Second: partial (matches www.google.com too)
- `contains` is case-sensitive

Exam Frequency: ★★☆☆ (High)

● **PROBLEM 33: Two Hosts Communication**

Question: Show traffic between 10.0.0.5 and 10.0.0.10

Display Filter:

```
ip.addr==10.0.0.5 and ip.addr==10.0.0.10
```

Explanation:

- Both IPs must be present
- One as source, other as destination
- Bidirectional

Exam Frequency: ★★★★★ (Very High)

● **PROBLEM 34: Multiple Ports (OR)**

Question: Show HTTP or HTTPS traffic

Display Filter:

```
tcp.port==80 or tcp.port==443
```

OR

```
tcp.port in {80 443}
```

Explanation:

- First: traditional OR

- Second: `in` operator (cleaner)
- Both work identically

Exam Frequency: ★★★★★ (Very High)

● PROBLEM 35: Exclude Traffic (NOT)

Question: Show all traffic EXCEPT from IP 192.168.1.100

Display Filter:

```
!(ip.addr==192.168.1.100)
```

OR

```
ip.addr!=192.168.1.100
```

Explanation:

- `!` = NOT operator
- `!=` = not equal
- Both work identically

Exam Frequency: ★★★★★ (High)

● PROBLEM 36: Subnet Filtering

Question: Show packets from network 172.16.0.0/16

Display Filter:

```
ip.src==172.16.0.0/16
```

Explanation:

- CIDR notation works in display filters
- `/16` = 172.16.x.x network
- Can use with src, dst, or addr

Exam Frequency: ★★★★★ (Very High)

● PROBLEM 37: Contains String

Question: Find packets containing "password" in payload

Display Filter:

frame contains "password"

OR (HTTP only):

http contains "password"

Explanation:

- `contains` searches packet data
- Case-sensitive
- Security analysis use case

Exam Frequency: ★★☆☆ (High)

● PROBLEM 38: Specific TCP Stream

Question: Show all packets in TCP stream number 5

Display Filter:

tcp.stream==5

Explanation:

- Each TCP connection has unique stream number
- Find number: Right-click → Follow → TCP Stream
- Isolates one conversation

Exam Frequency: ★★☆☆ (High)

● PROBLEM 39: Complex Combination

Question: Show HTTP to server 10.0.0.100 from subnet 192.168.1.0/24

Display Filter:

ip.src==192.168.1.0/24 and ip.dst==10.0.0.100 and tcp.port==80

Explanation:

- Three conditions with AND
- Directional (subnet to server)
- Specific protocol

Exam Frequency: ★★☆☆☆ (Very High)

● **PROBLEM 40: Multiple Conditions with Parentheses**

Question: Show web traffic (HTTP/HTTPS) between two hosts 10.0.0.5 and 10.0.0.10

Display Filter:

(ip.addr==10.0.0.5 and ip.addr==10.0.0.10) and (tcp.port==80 or tcp.port==443)

Explanation:

- **PARENTHESES ARE CRITICAL!**
- First group: identifies hosts
- Second group: identifies ports
- Both groups connected with AND

Exam Frequency: ★★☆☆☆ (Very High - Final Boss!)

🎯 **EXAM STRATEGY**

Must Memorize (Top Priority):

1. ☒ **Problem 1** - Basic host filtering
2. ☒ **Problem 4** - Two hosts communication
3. ☒ **Problem 6** - DNS queries (dst port 53) ← **VERY TRICKY!**
4. ☒ **Problem 7** - Multiple ports (HTTP/HTTPS)
5. ☒ **Problem 12** - Host + multiple ports **WITH PARENTHESES!**
6. ☒ **Problem 16** - TCP SYN packets
7. ☒ **Problem 17** - Multiple exclusions **WITH PARENTHESES!**
8. ☒ **Problem 20** - Complex capture filter
9. ☒ **Problem 28** - TCP SYN flag (display)
10. ☒ **Problem 29** - TCP SYN-ACK
11. ☒ **Problem 31** - DNS queries (response==0) ← **VERY TRICKY!**

12. **Problem 40** - Complex display filter

Common Exam Tricks:

-  **DNS queries vs responses** (dst port 53 vs src port 53)
-  **Parentheses placement** (changes entire logic!)
-  **Capture vs Display syntax** (host vs ip.addr)
-  **SYN vs SYN-ACK** (understanding 3-way handshake)
-  **Source vs Destination** (src vs dst, very important!)

 Quick Reference

Capture Filter Syntax:

```
host 10.0.0.1           # IP address
src/dst host 10.0.0.1    # Directional
net 192.168.1.0/24       # Network
port 80                  # Port
tcp port 80              # Protocol + Port
host 10.0.0.1 and tcp port 80 # Combination
(port 80 or port 443)    # Parentheses for OR
not tcp port 22           # Exclusion
tcp[13] & 2 != 0         # TCP flags (SYN)
```

Display Filter Syntax:

```
ip.addr==10.0.0.1       # IP address
ip.src==10.0.0.1        # Source
tcp.port==80             # Port
http                     # Protocol
http.request.method=="GET" # HTTP method
tcp.flags.syn==1         # TCP flag
dns.flags.response==0    # DNS query
tcp.stream==5            # TCP stream
frame contains "password" # String search
!(ip.addr==10.0.0.1)     # Exclusion
```

TCP Flags (MUST KNOW):

Display Filter	Capture Filter	Hex
-----	-----	---
tcp.flags.syn==1	tcp[13] & 2 != 0	0x02

tcp.flags.ack==1	tcp[13] & 16 != 0	0x10
tcp.flags==0x02	tcp[13] == 2	SYN
tcp.flags==0x12	tcp[13] == 18	SYN-ACK
tcp.flags==0x10	tcp[13] == 16	ACK

Common Ports (Memorize!):

20/21 = FTP
22 = SSH
23 = Telnet
25 = SMTP
53 = DNS
80 = HTTP
110 = POP3
143 = IMAP
443 = HTTPS
3389 = RDP
8080 = HTTP Alternate

⚠ CRITICAL MISTAKES TO AVOID

1. Parentheses Error (MOST COMMON!)

- ❌ WRONG: host 10.0.0.1 and port 80 or port 443
- ✅ RIGHT: host 10.0.0.1 and (port 80 or port 443)

Why wrong?

It means: (host AND port 80) OR (port 443 from ANYWHERE)

2. DNS Direction Confusion

- ❌ WRONG: udp port 53 (both queries and responses)
- ✅ RIGHT: udp dst port 53 (queries TO server only)
- ✅ RIGHT: udp src port 53 (responses FROM server only)

3. Capture vs Display Syntax

- ❌ WRONG in Capture: ip.addr==10.0.0.1
- ✅ RIGHT in Capture: host 10.0.0.1
- ❌ WRONG in Display: host 10.0.0.1
- ✅ RIGHT in Display: ip.addr==10.0.0.1

4. NOT Logic Error

✗ WRONG: not port 80 or port 443

✓ RIGHT: not (port 80 or port 443)

Why wrong?

First means: (NOT port 80) OR (port 443) - includes 443!

5. Equality Operator

✗ WRONG in Display: ip.addr=10.0.0.1 (single =)

✓ RIGHT in Display: ip.addr==10.0.0.1 (double ==)

PRACTICE DRILL (5 Minutes)

Answer WITHOUT looking at solutions:

1. Capture HTTP to server 10.0.0.1
2. Show DNS queries (display filter)
3. Capture traffic between 192.168.1.5 and 192.168.1.10
4. Show HTTP GET requests
5. Capture web traffic (HTTP/HTTPS) from subnet 10.0.0.0/24
6. Show TCP SYN packets
7. Capture all except SSH
8. Show packets from 172.16.0.0/16 to 10.0.0.0/8
9. Show HTTP 404 errors
10. Capture TCP SYN-ACK packets

Answers:

1. host 10.0.0.1 and tcp port 80
2. dns.flags.response==0
3. host 192.168.1.5 and host 192.168.1.10
4. http.request.method=="GET"
5. src net 10.0.0.0/24 and (tcp port 80 or tcp port 443)
6. tcp.flags.syn==1 and tcp.flags.ack==0
7. not tcp port 22
8. ip.src==172.16.0.0/16 and ip.dst==10.0.0.0/8
9. http.response.code==404
10. tcp.flags==0x12 OR tcp.flags.syn==1 and tcp.flags.ack==1

Score yourself:

- 10/10 = Ready for exam! 🎉
 - 7-9/10 = Almost there, review mistakes
 - <7/10 = Practice more!
-

FINAL CHECKLIST

Day Before Exam:

- ☐ Review all 40 problems once
- ☐ Memorize TCP flags (SYN, ACK, SYN-ACK)
- ☐ Memorize common ports (80, 443, 22, 53, 25)
- ☐ Practice parentheses placement
- ☐ Understand DNS dst vs src port 53
- ☐ Know difference: Capture vs Display syntax

During Exam:

- ☐ Read question carefully (queries vs responses?)
 - ☐ Check if asking for Capture or Display filter
 - ☐ Use parentheses for OR conditions
 - ☐ Verify src/dst direction
 - ☐ Double-check TCP flag questions
 - ☐ If complex, break into parts first
-

Good Luck!

Remember:

- These 40 problems cover 90% of exam questions
- Focus on problems marked ★★★★★
- Parentheses are your friend (and enemy if forgotten!)
- Practice makes perfect - do the drill multiple times

You've got this! 💪